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Applying Artificial Intelligence for Emotion Detection from Text Messages

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***Abstract*— Human interactions are usually fostered by emotions. There are numerous techniques to identify these emotions, such as through speech, body language, and facial expressions. To determine the emotions included in the text, text-based emotion detection is required. This paper develops Convolutional Neural Network (CNN) based emotion recognition in text. The objective of this paper is to evaluate the text and categories the many feelings it expresses including joy, fury, grief, surprise, love, and terror. The accuracy achieved with this method was 93.13%.**

Keywords—Emotion Detection, Machine Learning, Emotion Analysis.

I. Introduction

Emotions have an enormous impact in decision making and are an inherent aspect of human existence. Despite having been in use for some time, computers and other machines have historically depended on trustworthy information to make decisions. Humans are essentially capable of expressing their emotions through their faces, actions, words, and texts. These days, writings can be produced in a variety of ways, such as stories from the media, microblogs, and posts on social networking sites. From the verbiage of these posts, text mining may be used to identify and disclose a range of attributes, including emotions. Determining the feelings underlying these posts is a very difficult and time-consuming endeavour.

Comprehensive data sets can be reviewed by machine learning methods that are also capable of identifying patterns that people could be remiss. It usually improves the ever-growing volumes of data being handled with regards to accuracy and efficiency. Additionally, it permits immediate modification without requiring human involvement. The technique of completing a job by giving the system labelled information, and output precedents is more commonly referred to as supervised machine learning.

Emotional analysis may be useful in a variety of contexts. One such domain is human-computer interaction. Emotion recognition enables computers to make more effective judgements for people. This work is a extension of the author's previous paper titled "Emotion Detection and Recommender System using Machine Learning". Here's the layout for the rest of the paper. Section II examines the literature review and current approaches for emotion detection. Section III discusses our intended study using machine learning methodology. Section IV focuses on the findings. The paper is concluded in Section V.

II. Literature Survey

The authors of [1] describes a system that uses facial recognition to identify an individual's feelings and then adds some quotes to it. To categorise a commonality emotions, image processing and a neural network-based approach are used. Pre-processing and data methods of enhancement are used to increase training effectiveness and classification performance.

The authors of [2] uses a convolutional neural network (CNN) to compare word embeddings such as Word2Vec, BERT, and GloVe to produce emotion recognition in text. The best accuracy is produced by BERT+CNN, as stated by authors, when contrasted with Word2Vec+CNN and GloVe+CNN. This illustrates how successfully the CNN and BERT integrated algorithms work together to identify emotions in text.

The authors of [3] suggest A system of artificial intelligence that makes use of CNN's several layers to determine the emotions on a person's face. comprehensive analysis of comprehensive Face Expression Recognizer (FER), using techniques and datasets that illuminate these fundamental issues. The precision of emotion recognition throughout the training and testing phases is compared in the results.

The authors of [4] utilised an approach that enables the computer to instinctively detect the facial identification of human emotions with the assistance of CNN and OpenCV in order to recognise genuine emotions from an individual's facial gesture.

The authors of [5] have put out a hybrid model for text emotion identification that combines deep learning and machine learning. Deep learning techniques such as CNN and Bi-GRU were utilised. One method of machine learning is support vector machines. Sentences, tweets, and dialogues are the three types of datasets used to assess the effectiveness of the suggested strategy.

The authors of [6] provide students with an online space to share photos of the many the extracurricular pursuits they engage in. All of the students at an institution may use their portal as a portfolio. Students can use this gateway to find out about a range of events happening inside the institution.

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III. Proposed System

Emotion detection is a subfield of sentiment analysis where one investigates and extracts emotions. Convolutional Neural Networks (CNN) are used by the proposed system to identify emotions in text input by the user. CNN's approach to emotion recognition mostly leverages analysis to ascertain a person's emotional state.

Convolutional neural networks (CNNs) are a type of deep learning network architecture specifically designed to handle grid-like input. A convolutional neural network is composed of several layers, including the input layer, pooling layer, convolutional layer, and fully connected layers.

By implementing a focus mechanism, the proposed study enables CNN to solely focus on words that have a greater impact on identification or on the aspects among the traits that require more focus. Building the framework that recently acquired information for their users thoughts and monitoring social media are the main objectives of the task, since there is knowledge of the public's views beyond such topics.

Through Deep Learning, the system is able to understand sentence structure and contextual interdependency. First, an associated emotion dataset is constructed. The neural network receives this labelled dataset after which it is trained to interpret fresh input and become more accurate. A variety of alternatives, such as Recurrent Neural Network and Convolution Neural Network, are available for choosing training models. CNN is used in our proposed method. Pre-processing of the text is conducted prior to applying the algorithms to the input. This translation of the raw information into an alternative manner is simple and efficient to perform.

The collection includes dataset that express the six primary emotions. The dataset contains 20,000 instances of two features: text and emotion. Approximately 70% (14,000) of the 20,000 samples are employed in training, with the remaining thirty percent (6,000) for testing. During data pretreatment, the dataset is cleaned to remove unnecessary words, punctuation, and other elements. Figure 1 shows how the proposed system recognises and analyses emotions from a text or sentence input by the user. Figure 2 depicts the activity diagram.

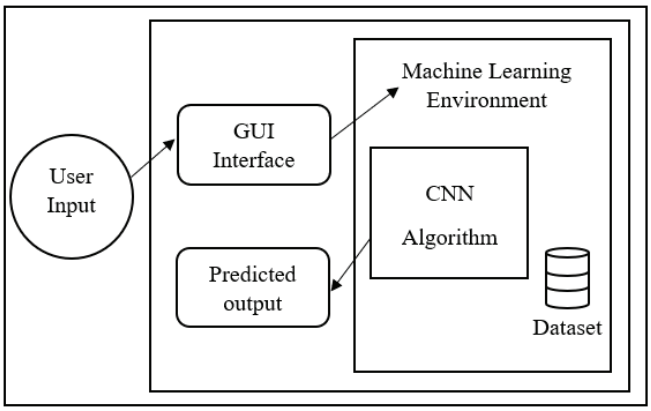

Fig. 1. System Architecture

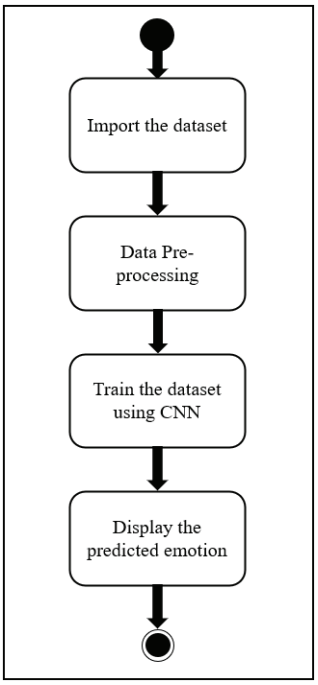

Fig. 2. Flowchart of Activity

The experience that users experience is crucial for a website to live up to the demands of users and continue performing effectively. A user interface is created in order to implement the algorithms for machine learning that have been developed. The user-provided information is then fed into a CNN algorithm, which trains it and outputs the appropriate emotion. Figure 3 and figure 4 shows the proposed work.

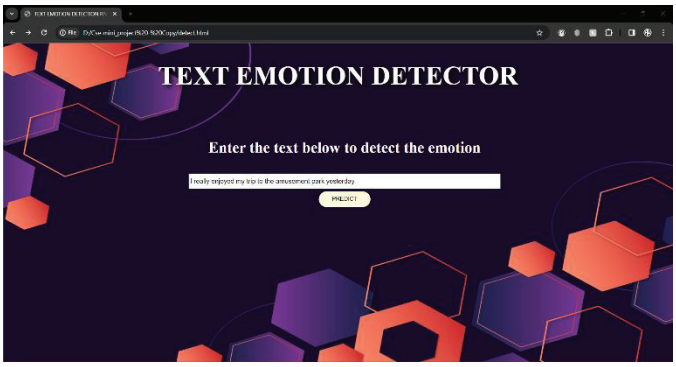

Fig. 3. Getting input from user

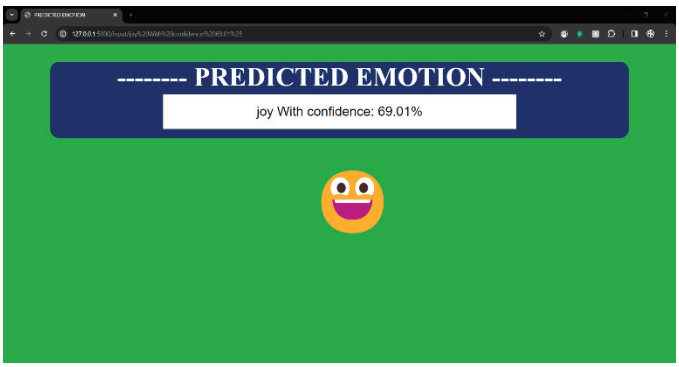

Fig. 4. Prediction results

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IV. Implementation and Result

Three kinds of metrics are used to assess the proposed work: the AUC-ROC curve, the classification report, and the confusion matrix.

Using a confusion matrix is one way to describe how well a categorization system is working. The table containing four unique sets of expected and actual values. Figure 5 display the CNN algorithm's confusion matrix.

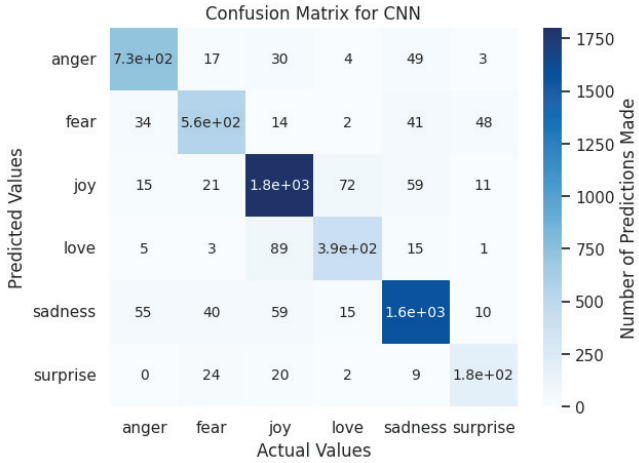

Fig. 5. Confusion matrix for CNN algorithm

A classification report is a machine learning assessment of performance data. The classification report for the CNN algorithm under study are illustrated in Table I.

TABLE I. Classification Report for CNN Algorithm

Emotions	Precision	Recall	F1-Score	Support
anger	0.92	0.93	0.93	507
Fear	0.91	0.96	0.93	831
Joy	0.96	0.95	0.96	1980
Love	0.94	0.89	0.91	697
Sadness	0.93	0.91	0.92	230
Surprise	0.94	0.94	0.94	1755
Accuracy			0.93	6000
Macro avg	0.93	0.93	0.93	6000
Weighted avg	0.93	0.93	0.93	6000

The outcome of the implementation shows the accuracy in detecting the emotions about 93.13% using the CNN algorithm.

At different criterion values, outcomes for classification tasks may be evaluated using the AUC-ROC curve. An overall performance metric across all potential categorization criteria is offered by AUC. The likelihood that the model values a random affirmative example higher than an arbitrary unfavourable example is one approach to analyse AUC.

In contrast, the greater the AUC, the more adept the model is in distinguishing between emotions. One emotion vs. rest state classification model is used to find the AUC-ROC curve for a classification that involves multiple classes issue, such as emotion detection in the suggested study. Figure 6 depicts the AUC-ROC curve for the CNN algorithm used in the proposed work.

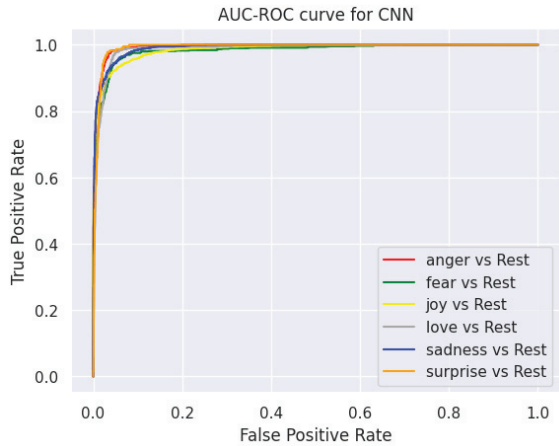

Fig. 6. AUC-ROC curve for CNN algorithm

V. Conclusion and Future Work

The study's usage of the Convolutional Neural Network (CNN) algorithm has demonstrated its effectiveness, as seen by its amazing 93% accuracy rate. This outcome highlights how robust and trustworthy CNNs are when processing complicated data, particularly with regards to text emotion recognition. Still, it is critical to recognise the continuous attempts to improve the speed, scalability, and interpretability of CNNs. To further push the limits of CNN capabilities, potential subjects for further research include innovative architectures, optimisation methods, and data augmentation approaches. In summary, the 93.13% accuracy rate attained is evidence of CNN effectiveness and highlights CNNs' critical role in pushing the boundaries of AI and machine learning research. We expect even more revolutionary breakthroughs as we explore the world of CNN, which will transform many fields and open the door to a future, especially those who are capable of detecting emotions.

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